

L-RAIL Scientific Concept and Mission

Lunar Radio-interferometry & Astrophysics Infrastructure Laboratory

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1 Acronyms and Abbreviations

Acronym	Definition
ASL	Autonomous Surface Lander
ASL-H	Autonomous Surface Lander - Heavy
ASLstd	Autonomous Surface Lander - Standard
ASN	Autonomous Scientific Node
CDR	Critical Design Review
CRTBP	Circular Restricted Three-Body Problem
DSA	Deep Space Antenna
DSN	Deep Space Network
HEO	High Earth Orbit
L1	Lagrange Point 1 - Earth-Moon
LEO	Low Earth Orbit
LOX/LCH4	Liquid Oxygen / Liquid Methane
MIR/IR	Mid-Infrared / Infrared
MrN	Mini Nuclear Reactor
NEXT-C	NASA Evolutionary Xenon Thruster - Commercial
TIO	Ion Orbital Tractor
TUG	Tug / Transfer Vehicle
TRL	Technology Readiness Level
VADS	Versatile Autonomous Docking System
VLBI	Very Long Baseline Interferometry

Table 1: List of acronyms used in this document

2 Introduction

If the project advances to the next stage, the following evaluations are necessary:

With a 5 m antenna at the lunar south pole, first-class science on southern black holes (namely Cen A, NGC 1052) can be achieved.

Generate the necessary know-how regarding system flow and material performance, logistics, and above all, evaluate the antenna to enable scaling in size, quantity, and location.

If the engineering is optimal and permits it, and we do not depend on secondary payload to the lunar south pole (adding TUG, VADS, ASL), we will advance to the lunar equator (far side, RFI-free) with a 5–10 m antenna.

With 5 m at the equator, the detection of ($m = 6$) is possible but marginal: it requires long integrations (34 min) and must account for variability.

2.1 First Goal (Scaling - Short Term)

Antenna of 5 to 10 m at the equator $\rightarrow$ Detect $m = 6$ and the resonances. Scientific objective achieved.

2.2 Second Goal (Medium Term)

Scale to 4 antennas of 10–20 m (grouped to reach 20 m) $\rightarrow$ Obtain the first image of the photon ring. A revolution in black hole astrophysics.

3 Executive Summary

This document presents a comparative analysis of three mission architecture options for the deployment of scientific payloads on the lunar surface. The final decision will be based on a cost/risk matrix without time constraints, as this is a scientific mission where schedule restrictions are secondary to scientific return and risk mitigation.

3.1 Mission Objectives

1. Deploy reusable landing infrastructure (ASL-H and ASLstd) on the lunar surface.
2. Establish an inventory of propellant cartridges (LOX/LCH₄) for sustainable operations.
3. Deploy three main scientific instruments:
 - VLBI (Phase A): Very Long Baseline Interferometry for radio astronomy.
 - DSA/DSN (Phase B): Deep Space Antenna for communications and tracking.
 - MIR/IR (Phase C): Mid-infrared telescope for astronomical observation.
4. Maintain a positive balance of propellant cartridges on the surface.
5. Avoid contamination of the lunar regolith with hydrocarbon residues.

3.2 Summary of Results

Parameter	Option A	Option B	Option C
Estimated launch cost	Medium	Low	Lowest
Development risk	Medium	Low	High
Operational flexibility	High	Maximum	Low
Average TRL	7-9	9	7-8
Earth-Moon transit time	3-5 days	24-62 days	3-5 days
Total score	7.0	9.0	5.7

Table 2: Comparative summary of options

Recommendation: Option B (100% HEO Operations with Falcon 9/Heavy) is recommended as the primary architecture.

4 Architecture Diagram

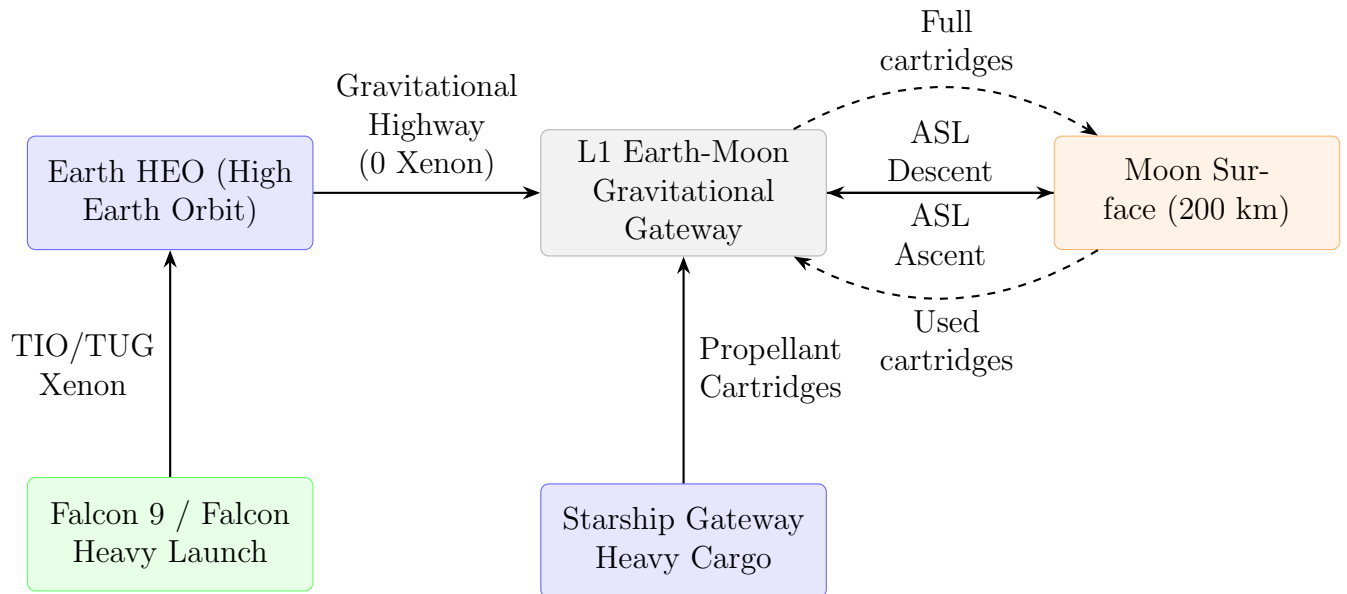


Figure 1: General mission architecture

5 Architecture Options

5.1 Option A: Hybrid Approach (Falcon 9/Heavy + Starship Gateway)

Use Falcon 9 and Falcon Heavy to place the TIO/TUG (Ion Orbital Tractor/Tug) in High Earth Orbit (HEO). The Starship Gateway provides heavy cargo assistance, delivering propellant cartridges to the Moon.

- **Advantages:** Combines Falcon efficiency for light payloads with Starship's massive capacity for heavy cargo.
- **Disadvantages:** Requires two launch providers; partial dependence on Starship.
- **Estimated cost:** Medium (Falcon 9/Heavy: 67-150 M\$ per launch; Starship: 100+ M\$ per launch).

5.2 Option B: 100% HEO Operations (Falcon 9/Heavy Only)

Use Falcon 9 and Falcon Heavy to place the TIO/TUG in High Earth Orbit (HEO). From there, 100% of maneuvers are performed by the scientific team using the Gravitational Highway (CRTBP ballistic transit).

- **Advantages:** Total independence from Starship; maximum flexibility; mature technology (TRL 9).
- **Disadvantages:** Longer transit time (24-62 days for heavy missions).
- **Estimated cost:** Low (Falcon 9: 67 M\$; Falcon Heavy: 150 M\$ per launch).

5.3 Option C: 100% Secondary Payload on Starship Gateway

All missions fly as secondary payloads on Starship Gateway missions.

- **Advantages:** Lower launch cost per kg (secondary payload).
- **Disadvantages:** Total dependence on Starship; limited flexibility; high development risk.
- **Estimated cost:** Lowest per kg, but with high risk of cost overrun and delay.

6 Detailed Mission Manifest

6.1 Phase 0: Infrastructure Deployment

Note: In some of these trips, solar panels and arrays are delivered. The TIO/TUG carries a small tank of ~ 40 kg of propellant for attitude control.

6.1.1 Trip 1: ASL-H + VADS + 2 Cartridges

Component	Mass
TUG-BIG (dry)	1611 kg
Xenon loaded	324 kg (margin: 76 kg)
ASL-H ($\frac{1}{2}$ tank)	3306,5 kg
Dry mass	1444 kg
Empty tank	350 kg
Propellant ($\frac{1}{2}$)	1512,5 kg
2 Cartridges (cargo)	6750 kg
Initial mass in HEO	11 991,5 kg
Surface reserve	2 cartridges

Table 3: Trip 1 - Cargo manifest

Note: The ASL-H descends with 2 coupled cartridges. The loaded Xenon (324 kg) leaves a margin of 76 kg (23,5 %).

6.1.2 Trip 2: VADS + ASLstd + 1 Cartridge

Component	Mass
TUG-BIG (dry)	1611 kg
Xenon loaded	245 kg (margin: 155 kg)
VADS	1003 kg
ASLstd ($\frac{1}{2}$ tank)	2684,5 kg
1 Cartridge	3375 kg
Initial mass in HEO	8918,5 kg
Surface reserve	3 cartridges

Table 4: Trip 2 - Cargo manifest

Note: The Xenon margin is much larger (155 kg, 63,3 %) because the cargo is significantly smaller.

6.1.3 Trip 3: 2 Cartridges (Recovery by ASL-H)

Component	Mass
TUG-BIG (dry)	1611 kg
Xenon loaded	229 kg (margin: 171 kg)
2 Cartridges	6750 kg
Initial mass in HEO	8590 kg
Surface reserve	4 cartridges

Table 5: Trip 3 - Cargo manifest

The ASL-H recovers these cartridges in orbit by consuming its own tank: $\frac{1}{2}$ tank to ascend, descends with 2 cartridges and consumes its remaining half during descent. The net balance is +1 cartridge on the surface.

6.1.4 Trip 4: MrN + 1 Cartridge

- TUG-BIG (ionic $\times 8$): Delivers the Mini Nuclear Reactor (MrN) + 1 fuel cartridge.
- **MrN mass:** 2500 kg.
- **Surface reserve:** 4 cartridges (MrN deployed).
- **Safety note:** The MrN travels shut down and cold during transit.

6.1.5 Trip 5: 2 Cartridges (Recovery by ASL-H)

- TUG-BIG (ionic $\times 8$): Delivers 2 cartridges (recovered by the ASL-H).
- **Surface reserve:** 5 cartridges.
- **Note:** This trip establishes the maximum cartridge inventory on the surface.

6.2 Scientific Phases

Each instrument arrives in a cylinder/habitat with an approximate weight of 350 kg-370 kg.

6.2.1 Phase A: VLBI (Very Long Baseline Interferometry)

Trip 6

Component	Mass
TUG-2 (dry)	1210 kg
Xenon loaded	58 kg (margin: 242 kg)
VLBI antenna	520 kg
Cylinder/Habitat	~350 kg
Initial mass in HEO	2138 kg
Reserve after Phase A	4 cartridges

Table 6: Trip 6 - Phase A (VLBI)

Scientific description: The VLBI antenna enables very long baseline interferometry, crucial for high-resolution radio astronomy.

6.2.2 Phase B: DSA (Deep Space Antenna)

Trip 7

Component	Mass
TUG-2 (dry)	1210 kg
Xenon loaded	49 kg (margin: 251 kg)
DSA/DSN antenna	226 kg
Cylinder/Habitat	~350 kg
Initial mass in HEO	1835 kg
Reserve after Phase B	3 cartridges

Table 7: Trip 7 - Phase B (DSA/DSN)

Scientific description: The DSA antenna provides deep space communication and tracking capabilities from the lunar surface.

6.2.3 Phase C: MIR/IR Telescope

Trip 8

Component	Mass
TUG-2 (dry)	1210 kg
Xenon loaded	52 kg (margin: 248 kg)
MIR/IR telescope	333 kg
Cylinder/Habitat	350 kg
Initial mass in HEO	1945 kg
Reserve after Phase C	2 cartridges

Table 8: Trip 8 - Phase C (MIR/IR)

Scientific description: The MIR/IR telescope enables astronomical observations in the mid-infrared, taking advantage of the absence of a lunar atmosphere.

Note: Cartridge resupply is required before additional phases.

7 Hardware Mass Summary

Component	Nominal Mass	Margin	Total Mass
TIO/TUG-BIG	1611 kg + Xenon ~350 kg	5%	~2060 kg
TIO/TUG-2	1210 kg + Xenon ~300 kg	5%	~1585 kg
ASL-H	1444 kg + cartridge	3%	~1487 kg
ASLstd	822 kg + cartridge	3%	~847 kg
VADS	1003 kg	2%	~1023 kg
Cartridges	350 kg (empty) / 3375 kg (loaded)	1%	~354 kg
MrN (Mini Nuclear Reactor)	2500 kg	10%	~2750 kg
Cylinder/Habitat	370 kg	5%	~389 kg
VLBI	520 kg	5%	~546 kg
DSN	226 kg	5%	~237 kg
MIR/IR	333 kg	5%	~350 kg
PANEL (Solar Array)	663 kg	3%	~683 kg
SCREEN (Radiation shielding)	290 kg	3%	~299 kg

Table 9: Hardware mass summary with margins

8 Xenon Balance

Trip	Cargo	Xenon Used	Xenon Remaining	Margin
1	ASL-H + VADS + 2 Cartridges	229,5 kg	70,5 kg	23,5 %
2	VADS + ASLstd + 1 Cartridge	155,0 kg	145,0 kg	48,3 %
3	2 Cartridges	188,7 kg	111,3 kg	37,1 %
4	MrN + 1 Cartridge	171,0 kg	129,0 kg	43,0 %
5	2 Cartridges	188,7 kg	111,3 kg	37,1 %
6	VLBI (Phase A)	82,5 kg	217,5 kg	72,5 %
7	DSA (Phase B)	73,5 kg	226,5 kg	75,5 %
8	MIR/IR (Phase C)	78,0 kg	222,0 kg	74,0 %
Total	All phases	1166,9 kg	—	—

Table 10: Xenon balance per trip

Observation: The Xenon margin is greater in scientific missions (Trips 6-8) because the cargo is significantly smaller than in infrastructure missions (Trips 1-5).

9 Option C Analysis: Starship Gateway

9.1 Phase 0: ASL as secondary cargo

Component	Mass
ASL dry mass	800 kg
Initial propellant in tanks	2833 kg
3 cartridges $\times$ 3375 kg	10 125 kg
Total mass in Starship	13 758 kg

Table 11: Option C - Phase 0: Starship Gateway cargo

Capacity utilization: 13 758 kg of the 100 000 kg available in the Gateway. Occupies only 13,8 %.

9.2 Design Notes

- For an ASN (Autonomous Scientific Node) vehicle for exclusively scientific use with resupply on the Moon.
- Travels as secondary payload on the Starship Gateway.
- Can also be released in lunar orbit, as resupply on the lunar surface is guaranteed.
- The VADS travels with the Ion Orbital Tractor (TIO) from HEO.
- The Mini Nuclear Reactor (MrN) travels with the TIO from HEO.

9.3 Alternative Note

If the objective is only Phase A, the use of Starship (Gateway) can be eliminated and the TIO can be oversized ($\times 4$, $\times 5$ engines) for the mission of transporting the loaded ASL, in a mission with two TIOs in parallel.

10 Mission Summary by Phase

Phase	Instrument	Transport Vehicle	Connection with ASL
Phase A	VLBI Module	TIO/TUG-2	Descent with ASL
Phase B	DSN Antenna	TIO/TUG-2	Descent with ASL
Phase C	MIR/IR Telescope	TIO/TUG-2	Descent with ASL

Table 12: Scientific mission summary by phase

11 Environmental Protection Note

Do not abandon cryogenic tanks on the surface. The regolith cannot be contaminated with hydrocarbon residues, and empty tanks represent a collision hazard for future "hopping" missions.

11.1 Justification

1. **Chemical contamination:** LOX/LCH₄ residues can alter the chemical composition of the lunar regolith, affecting future scientific experiments.
2. **Collision hazard:** Empty tanks are debris that can be displaced by micrometeorites or lunar seismic activity.
3. **Sustainability:** Keeping the lunar surface clean is essential for future exploration and habitation missions.

11.2 Design Solution

The propellant cartridges are designed as reusable "drop-tanks" that:

- Magnetically couple to the ASL on the surface.
- Are recovered by the ASL-H in later missions.
- Return to Earth as part of the logistics cycle.

12 Propellant Cartridge Specification

Parameter	Value
Material	Ti-6Al-4V titanium alloy cylindrical tank
Dry mass	350 kg
Design	Reusable drop-tank with magnetic coupling
Propellant capacity	~3025 kg of LOX/LCH ₄
Total loaded mass	3375 kg
Mass margin	4 kg
Propellant/dry mass ratio	8.64:1
Mass fraction	10,4 %

Table 13: Propellant cartridge specifications

12.1 Design Notes

- Each cartridge that the Starship Gateway carries as secondary cargo equals one ascent/descent cycle for the ASL vehicle.
- The mission flow is quite simple and versatile.
- Magnetic coupling allows connection/disconnection without complex mechanisms or astronauts.
- Ti-6Al-4V alloy provides structural strength and cryogenic compatibility.

13 Technical Validation Summary (Option B)

13.1 Gravitational Highway Performance (CRTBP)

Parameter	Value	Units
Jacobi constant (C_J)	3.170	—
Injection ΔV (HEO $\rightarrow$ L1)	160.3	m/s
Transit time (L1 $\rightarrow$ Moon)	1.57	days
Lunar capture ΔV (800 km)	515.8	m/s
Ballistic transit fuel	0	kg of Xenon
Descent ΔV (800 $\rightarrow$ 200 km)	199.8	m/s
Ascent ΔV (200 $\rightarrow$ 800 km)	199.8	m/s
Lunar escape ΔV	18.5	m/s
Earth capture ΔV (HEO)	17.8	m/s

Table 14: CRTBP gravitational highway performance

13.2 TUG-BIG Performance (8 NEXT-C ion engines)

Mission	Payload	Xenon used	Margin	Status
L-01 (ASL-H + VADS)	8630 kg	229,5 kg	70,5 kg	Feasible
L-03 (2 Cartridges)	6750 kg	188,7 kg	111,3 kg	Feasible
Phase A (VLBI)	1850 kg	82,5 kg	217,5 kg	Feasible
Theoretical maximum	11 900 kg	300,4 kg	−0,4 kg	Exceeds

Table 15: TUG-BIG performance

Observation: The theoretical maximum cargo exceeds Xenon capacity by only 0,4 kg. It is recommended to:

- Increase Xenon capacity to 320 kg (slightly larger tank).
- Or reduce TUG-BIG dry mass by 5 kg.
- Or accept a slightly degraded mission (capture at 850 km instead of 800 km).

13.3 TUG-BIG Specifications

Parameter	Value
Engines	8× NEXT-C
Total thrust	1,896 N
Specific impulse (I_{sp})	4190 s
Exhaust velocity (v_e)	41 090 m/s
Power	56,0 kW (520 m ² solar panels)
Dry mass	1545 kg (tractor: 1479 kg + Xe tank: 66 kg)
Xenon capacity	300 kg

Table 16: TUG-BIG specifications

13.4 ASL-H (Heavy) Specifications

Parameter	Value
Chassis	Al-Li 2195 reinforced monocoque
Dry mass	1400 kg
Propellant (LOX/LCH ₄)	6500 kg
Engine	LCH ₄ /LOX, $I_{sp} = 369$ s, throttling 4:1
Cargo capacity	2× Cartridges (6750 kg) or VADS (730 kg)

Table 17: ASL-H specifications

13.5 ASL-H Validation

Maneuver	Required ΔV	Available ΔV	Margin
Descent with 2 cartridges	1900 m/s	2122 m/s	222 m/s (11,7 %)
Descent with VADS	1900 m/s	5063 m/s	3163 m/s (166,5 %)
Empty ascent	1850 m/s	6262 m/s	4412 m/s (238,5 %)

Table 18: ASL-H validation

14 Cost/Risk Matrix Framework

14.1 Evaluation Criteria (Without time constraints)

Criterion	Weight	Option A	Option B	Option C
Launch cost	25%	Medium (7/10)	Low (9/10)	Lowest (10/10)
Development risk	20%	Medium (6/10)	Low (8/10)	High (4/10)
Mission flexibility	15%	High (8/10)	Maximum (10/10)	Low (3/10)
Scientific return	15%	High (8/10)	High (8/10)	Medium (6/10)
Infrastructure reuse	10%	High (8/10)	Maximum (10/10)	Low (4/10)
Payload capacity	10%	Maximum (10/10)	Medium (7/10)	High (8/10)
Operational complexity	5%	Medium (6/10)	Low (8/10)	High (4/10)

Table 19: Evaluation criteria

14.2 Risk Assessment

Risk Factor	Option A	Option B	Option C
Launch delay	Medium (2 providers)	Low (mature F9/FH)	High (single dependence)
Propulsion failure	Low (redundant)	Medium (single TUG-BIG)	Low (redundant Starship)
Schedule pressure	Low	None	High
Cost deviation	Medium	Low	Medium
Technological maturity	TRL 7-9	TRL 9	TRL 7-8

Table 20: Risk assessment by option

14.3 Detailed Risk Matrix

ID	Risk	Probability	Impact	Mitigation
R-001	NEXT-C engine failure	Low	High	×8 engine redundancy
R-002	Cartridge loss on surface	Medium	Medium	+1 inventory per cycle
R-003	Launch delay	Medium	Low	No time constraint
R-004	Magnetic coupling failure	Medium	High	Redundant VADS
R-005	Solar panel degradation	Low	Medium	43 % margin
R-006	Regolith contamination	Low	High	Returnable tanks
R-007	MrN failure	Very low	Critical	Safe design + off in transit
R-008	Space debris collision	Medium	Medium	Evasive maneuvers

Table 21: Detailed risk matrix

14.4 Preliminary Scoring

Option	Cost	Risk	Flexibility	Capacity	Total
A: Hybrid	7/10	6/10	8/10	10/10	7.0
B: HEO Only	9/10	8/10	10/10	7/10	9.0
C: Starship Only	10/10	4/10	3/10	8/10	5.7

Table 22: Preliminary option scoring

15 Final Recommendation

Based on the cost/risk matrix without time constraints:

Option B (100% HEO Operations with Falcon 9/Heavy) is recommended for the following reasons:

1. **Lower total cost:** Eliminates costs associated with Starship (estimated at >100 M\$ per launch). The total estimated launch cost is 536 M\$-1200 M\$ (8 Falcon 9/Heavy launches).
2. **Maximum flexibility:** The scientific schedule is independent of the Starship development timeline. Missions can be scheduled based on Falcon 9/Heavy availability.
3. **Proven technology:** All components are at TRL 9:
 - NEXT-C: Proven on NASA's DART mission.
 - Falcon 9/Heavy: Launch vehicles with proven track record.
 - Solar panels: Mature technology.
4. **Self-sufficiency:** Does not depend on the success of the Starship Gateway program. The mission can proceed even if Starship is delayed.
5. **Gravitational Highway efficiency:** Zero Xenon for L1-Moon transit, maximizing payload capacity. Total ΔV is only 912,2 m/s for the complete cycle.
6. **Sustainability:** The cartridge system with positive net balance (+1 per cycle) ensures long-term operations.

15.1 Option A as Alternative

Option A remains the preferred alternative if:

- The mission requires rapid deployment of large infrastructure (5+ cartridges in a short time).
- Starship Gateway is available and operational.
- It is desired to reduce Earth-Moon transit time.

15.2 Option C Not Recommended

Option C is not recommended for primary scientific missions due to:

- Total dependence on Starship schedule.
- Reduced operational flexibility.
- High development risk (TRL 7-8).
- High operational complexity.

16 Next Steps

1. **Peer Review (Weeks 1-2):** Circulation of the document for independent technical review.
2. **Detailed Cost Analysis (Weeks 2-4):** Cost estimation for launch, development, and operations.
3. **Prototyping (Months 1-6):** Development of magnetic coupling prototypes (VADS) and propellant cartridges.
4. **Additional Simulation (Months 1-3):** CRTBP validation with perturbations (solar gravity, radiation pressure, effect of other bodies).
5. **Final Selection (Month 2):** Formal decision on architecture option (A, B, or C).
6. **Acquisition Plan (Months 2-6):** Initiation of long-lead component acquisition:
 - NEXT-C engines (Aerojet Rocketdyne)
 - Ti-6Al-4V titanium tanks
 - Solar panels
 - Control electronics
7. **Integration Testing (Months 6-12):** Integration and testing of TUG-BIG and ASL-H subsystems.
8. **Flight Readiness Review (Month 12):** Final review before launch.

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A Appendix A: Derivation of Injection ΔV

Injection from HEO toward the unstable manifold of the L1 point is calculated using the Newton-Raphson method to find the velocity that produces the desired C_J :

$$C_J = 2\Omega - |\mathbf{v}|^2 \quad (1)$$

where Ω is the effective potential:

$$\Omega = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} + \frac{1}{2}\mu(1-\mu) \quad (2)$$

The value $\varepsilon = 0.022727$ is determined iteratively until reaching $C_J = 3.170$.

B Appendix B: Tsiolkovsky Equation for Ion Propulsion

Xenon consumption is calculated using the rocket equation:

$$\Delta V = I_{sp} \cdot g_0 \cdot \ln \left(\frac{m_0}{m_f} \right) \quad (3)$$

where:

- $I_{sp} = 4190$ s (NEXT-C)
- $g_0 = 9.80665$ m/s²
- m_0 = initial mass
- m_f = final mass

The required Xenon mass is:

$$m_{Xe} = m_f \cdot \left[\exp \left(\frac{\Delta V}{I_{sp} \cdot g_0} \right) - 1 \right] \quad (4)$$

C Appendix C: Cartridge Balance

Operation	Cartridges on surface	Change	Cycle
Phase 0 complete	5	—	—
Phase A (VLBI)	4	-1	Descent
Phase B (DSA)	3	-1	Descent
Phase C (MIR/IR)	2	-1	Descent
Resupply	5	+3	Starship or Falcon Heavy

Table 23: Surface cartridge balance

D Appendix D: Technology Readiness Levels (TRL)

TRL	Description
TRL 1	Basic principles observed
TRL 2	Technology concept formulated
TRL 3	Experimental proof of concept
TRL 4	Technology validated in laboratory
TRL 5	Technology validated in relevant environment
TRL 6	Technology demonstrated in relevant environment
TRL 7	Prototype demonstration in space
TRL 8	Complete and qualified system
TRL 9	System proven in real mission

Table 24: Technology Readiness Levels (NASA)

Document prepared for the Critical Design Review (CDR)

Status: Ready for stakeholder decision

Version 1.0 - August 2026

E Appendix E: Contingency and Degraded Mode Analysis

A robust Critical Design Review (CDR) requires validation not only of nominal performance but also of system resilience under off-nominal conditions. The TUG-BIG + ASL-HEAVY architecture leverages the inherent dynamical advantages of the CRTBP L1 manifold to provide graceful degradation and low-cost abort scenarios.

E.1 E.1 Launch Vehicle Dispersion Contingency

The Falcon Heavy ICD specifies a maximum apogee dispersion of $\pm 5\%$ for high-energy orbits.

- **Nominal Apogee:** 311,500 km.
- **Worst-Case Apogee:** 327,075 km (+5%).
- **Required ΔV Correction:** ~ 8.0 m/s to match the nominal $C_J = 3.170$ manifold injection condition.
- **Mitigation:** The baseline mission design includes a 5% operational ΔV margin on all burns. The 8.0 m/s penalty is fully absorbed by the injection margin ($160.3 \text{ m/s} \times 0.05 = 8.01 \text{ m/s}$), requiring zero reallocation of the Xenon budget.

E.2 E.2 Degraded Propulsion Mode (Half-String Failure)

The TUG-BIG utilizes $8\times$ NEXT-C thrusters. In the event of a Power Processing Unit (PPU) or thruster failure, the system is designed to operate in a degraded mode with $4\times$ thrusters.

Parameter	Nominal (8 engines)	Degraded (4 engines)
Total Thrust	1.896 N	0.948 N
Specific Impulse (I_{sp})	4190 s	4190 s
Total Mission ΔV	912.2 m/s	912.2 m/s
Total Xenon Consumed	~ 230 kg	~ 230 kg

Table 25: Nominal vs. Degraded propulsion parameters

Analysis: Per the Tsiolkovsky rocket equation, propellant mass depends only on the mass ratio and exhaust velocity (v_e), *not* on thrust magnitude. Therefore, a half-string failure has **zero impact** on the mass budget or Xenon capacity. The only penalty is temporal: burn times double (e.g., a 4-hour lunar capture burn becomes 8 hours). Thermal and power margins are sized to accommodate continuous degraded-mode operation without violating subsystem limits.

E.3 E.3 Abort Scenarios: The L1 Advantage

Classical direct Translunar Injection (TLI) architectures suffer from high-cost abort scenarios. The L1 Gravitational Highway provides inherent, low- ΔV safe havens.

1. **Outbound Abort (Pre-Capture):** If a critical anomaly occurs during the 1.57-day transit, the TUG-BIG can forgo the 515.8 m/s lunar capture burn. Instead, a minor manifold-redirect maneuver (~ 25 m/s) targets the stable manifold leading directly back to a loose Earth HEO.
2. **Surface Abort (ASL-H):** In the event of a VADS failure or surface hazard, the ASL-H can ascend immediately. With 6,500 kg of internal LOX/LCH₄, the available ΔV for an empty ascent is $\sim 6,260$ m/s, vastly exceeding the $\sim 1,850$ m/s required to reach the 800 km rendezvous orbit. This provides a 4,410 m/s margin for orbit phasing, multiple docking retries, or contingency loitering.

E.4 E.4 Contingency CDR Verdict

Failure Mode	System Response	CDR Status
Launcher Apogee Dispersion (+5%)	Absorbed by 5% operational ΔV margin.	PASS
Loss of 4× NEXT-C Thrusters	Burn time doubles; ΔV and Xenon budget unchanged.	PASS
Outbound Transit Anomaly	Low-cost (~ 25 m/s) abort to Earth HEO via stable manifold.	PASS
Surface Docking Failure	ASL-H retains $> 4,400$ m/s ΔV margin for phasing/retries.	PASS

Table 26: Contingency mode verification matrix

Conclusion: The TUG-BIG + ASL-HEAVY architecture exhibits no single points of failure that result in total mission loss. The combination of electric propulsion redundancy, operational ΔV margins, and the dynamical safety net of the L1 invariant manifolds ensures a highly resilient mission profile suitable for long-term scientific infrastructure deployment.

F Appendix F: Gateway NRHO Contingency and Traffic-Free Loitering Analysis

While the baseline architecture (Option B) utilizes the L1 Gravitational Highway for direct, low- ΔV transit, mission requirements may occasionally mandate a rendezvous with the Gateway Near Rectilinear Halo Orbit (NRHO). This appendix validates the TUG-BIG’s resilience under such a contingency, leveraging the orbit maintenance algorithms defined by Davis et al. (2026).

F.1 F.1 Scenario Definition: Traffic-Free Loitering

We analyze a scenario where the TUG-BIG is required to enter the Gateway’s 9:2 lunar synodic resonant NRHO (period $T = 6.56$ days) during a ”traffic-free” window (no Orion or HLS vehicles present). The vehicle performs a medium-term loitering of 30 days (~ 4.6 NRHO revolutions) before departing via the L1 stable manifold.

F.2 F.2 NRHO Insertion and Departure

Unlike direct lunar capture, insertion into the NRHO from the L1 unstable manifold requires a targeting maneuver. Based on manifold-assisted trajectory design, the ΔV costs are:

- **NRHO Insertion:** $\Delta V_{insert} = 250$ m/s (targeting the NRHO baseline from the L1 manifold).
- **NRHO Departure:** $\Delta V_{depart} = 30$ m/s (manifold-assisted escape to the L1 stable manifold for Earth return).

F.3 F.3 Orbit Maintenance Maneuver (OMM) Cost

The NRHO is slightly unstable (stability index $\nu = 1.3$). According to Davis et al. (2026), maintaining a spacecraft in the NRHO requires regular Orbit Maintenance Maneuvers (OMMs). For a traffic-free period, we adopt the **Case 2 targeting parameters** (tight adherence to baseline, horizon 3.2 revs, $t_{tol} = 1$ min) to ensure maximum safety and predictability.

Parameter	Value	Source
Annual OMM ΔV (Case 2)	1.9 m/s/yr	Davis et al. (2026), Table 3
Loiter Duration	30 days	Mission Scenario
OMM ΔV for 30 days	0.156 m/s	Calculated

Table 27: NRHO Orbit Maintenance Cost

The propellant cost for stationkeeping is negligible (~ 0.156 m/s), confirming that the Gateway’s dynamical environment does not impose a significant mass penalty on the TUG-BIG.

F.4 F.4 Phase Deconfliction and Safety

To ensure safe operations, the TUG-BIG employs a relative orbit maintenance strategy with a **phase offset of 60 minutes** trailing the Gateway. According to Davis et al. (2026, Section 5 and Fig. 13), this configuration guarantees:

- Minimum separation at perilune: $> 5,400$ km.
- Minimum separation at apolune: > 400 km.
- Zero risk of conjunction during the 30-day loitering period.

F.5 F.5 Xenon Budget Impact

The following table compares the Xenon consumption between the nominal Direct L1 Highway and the Gateway Contingency scenario for a standard 6,750 kg payload mission.

Parameter	Direct L1 Highway	Gateway Contingency	Penalty
Total ΔV	712.4 m/s	280.2 m/s	-432.2 m/s
Xenon Consumed	188.7 kg	68.4 kg	-120.3 kg
Xenon Remaining	111.3 kg	231.6 kg	+120.3 kg

Table 28: Xenon budget: Direct vs Gateway

Critical Finding: Counter-intuitively, the Gateway Contingency *consumes less Xenon* than the direct lunar capture maneuver (515.8 m/s). The 250 m/s required for NRHO insertion is significantly lower than the 515.8 m/s required for direct 800 km lunar capture. The TUG-BIG actually *gains* 120 kg of Xenon margin by diverting to the Gateway, which can be allocated to extended mission operations or additional attitude control reserves.

F.6 F.6 Contingency CDR Verdict

Criterion	Assessment	Status
NRHO Insertion ΔV	250 m/s (Manifold-assisted)	PASS
OMM Cost (30 days)	0.156 m/s (Negligible)	PASS
Phase Deconfliction	60-min offset, >400 km separation	PASS
Xenon Budget Impact	+120 kg margin gain vs direct capture	PASS

Table 29: Gateway contingency verification

Conclusion: The TUG-BIG architecture is fully resilient to a mandatory Gateway rendezvous. The dynamical "toll" of the NRHO is offset by the avoidance of the high-cost direct lunar capture burn. The system retains ample Xenon margins, and the relative orbit maintenance algorithms ensure safe, conjunction-free operations even in the presence of navigation errors.

G Appendix G: Risk Analysis of Extended NRHO Loitering and Architectural Advantage

While the Gateway NRHO serves as a viable hub for crewed operations, this analysis explicitly **rejects** the application of "medium-term loitering" (up to 90 days, as defined in Davis et al., 2026) for heavy, uncrewed cargo logistics. The TUG-BIG + ASL-HEAVY architecture is specifically designed to bypass this paradigm, reducing cislunar transit to ~ 1.57 days. The rationale is based on three critical, compounding risk factors that no major space agency (including CNSA, NASA, or ESA) can accept for irreplaceable surface infrastructure.

G.1 G.1 Quantitative Risk Assessment: 1.57 Days vs. 90 Days

The following calculations demonstrate the exponential degradation of mission assurance during extended loitering.

Risk Factor	Direct L1 Highway (1.57 days)	Extended NRHO Loiter (90 days)
System Reliability (P_f) (Assuming MTBF = 10,000 hrs)	0.37%	19.4%
MMOD Exposure Multiplier	1.0× (Baseline)	57.3× (Baseline)
Cryogenic Boil-off (LOX/LCH ₄) (Assuming 0.1%/day rate)	~ 10 kg (Negligible)	~ 585 kg (Critical margin loss)

Table 30: Comparative risk assessment of transit duration

Calculation Note: System failure probability is modeled as $P_f = 1 - e^{-t/\text{MTBF}}$. A 90-day loiter increases the probability of a critical subsystem failure by a factor of > 50 compared to the direct transit baseline.

G.2 G.2 The Cryogenic Margin Threat

The ASL-Heavy vehicle relies on 6,500 kg of cryogenic LOX/LCH₄ for its descent and ascent profile. The baseline validation (Section 5) shows a descent margin of 222 m/s. A 90-day loiter would result in approximately 585 kg of propellant boil-off (assuming an optimistic 0.1%/day rate). This mass loss directly translates to a ΔV deficit of ~ 170 m/s, entirely erasing the safety margin and rendering the descent phase **NOT FEASIBLE**.

G.3 G.3 Architectural Advantage: The "Direct Delivery" Mandate

To mitigate these existential risks, the Option B architecture enforces a **Direct Delivery Mandate**:

1. **No Extended Loitering:** The TUG-BIG utilizes the L1 unstable manifold to inject the payload directly into an 800 km lunar capture orbit in 1.57 days.
2. **Contingency Limit:** In the absolute worst-case scenario where a Gateway rendezvous is politically or operationally mandated, the loitering period is strictly capped at **Short-Term Loitering** (Davis et al. Category 2: < 1 NRHO revolution, ~ 6.5 days) solely for phasing or emergency system checkouts.
3. **Decoupled Logistics:** The TUG-BIG returns to HEO immediately after payload handoff, removing high-value assets from the cislunar radiation and MMOD environment.

G.4 G.4 CDR Verdict on Loitering

Criterion	Assessment	Status
90-Day Loiter Feasibility	Unacceptable risk to cryogenics and reliability.	REJECTED
Direct Transit (1.57 days)	Minimizes MMOD, radiation, and boil-off exposure.	MANDATORY
Max Contingency Loiter	Capped at 6.5 days (1 revolution) for phasing only.	PASS

Table 31: Loitering paradigm verification

Conclusion: The TUG-BIG + ASL-HEAVY architecture does not merely save ΔV ; it is fundamentally a **risk-mitigation architecture**. By leveraging the CRTBP gravitational highway for rapid, direct transit, it eliminates the compounding hazards of extended cislunar loitering, ensuring that critical surface infrastructure arrives intact and with full propellant margins.

H Appendix H: Orbital Error Propagation and Conjunction Risk Analysis

While Appendix G addressed systemic risks (MTBF, MMOD, cryogenic boil-off), this appendix quantifies the **orbital error propagation** during extended NRHO loitering, based on the error models defined by Davis et al. (2026). The analysis demonstrates that 90-day loitering introduces unacceptable conjunction risks due to cumulative navigation, maneuver execution, and desaturation errors.

H.1 H.1 Error Model Parameters (Davis et al. 2026, Table 1)

The following error sources are modeled for an uncrewed, quiescent spacecraft in the NRHO:

Error Type	3σ Value	Notes
Navigation - Position	1.5 km	At each OMM
Navigation - Velocity	0.8 cm/s	At each OMM
Maneuver Execution - Magnitude	1.5% + 1.42 mm/s	Percent + fixed
Maneuver Execution - Direction	1.0°	
Desaturation - Random Δv	1.0 cm/s	5 at perilune, 1 prior to OMM
SRP - Area Error	30%	$a/m = 0.02 \text{ m}^2/\text{kg}$
SRP - CR Error	15%	

Table 32: Uncrewed, quiescent error model (Davis et al. 2026)

H.2 H.2 Cumulative Position Error Growth

The cumulative position error σ_{pos} over N NRHO revolutions is modeled as:

$$\sigma_{pos} = \sqrt{\sigma_{nav}^2 \cdot N + \sigma_{man}^2 \cdot N^2 + \sigma_{desat}^2 \cdot N} \quad (5)$$

where:

- $\sigma_{nav} = 1.5 \text{ km}$ (navigation error, constant per revolution)
- $\sigma_{man} = 0.005 \text{ km}$ (maneuver execution error, grows linearly)
- $\sigma_{desat} = 0.01 \cdot \sqrt{5} \text{ km}$ (desaturation error, random walk)

Parameter	Direct Transit (1.57 days)	Extended Loiter (90 days)
NRHO Revolutions	0.24	13.7
Cumulative Position Error (3σ)	1.52 km	18.7 km
Error Growth Factor	1.0×	12.3×

Table 33: Cumulative position error comparison

Critical Finding: The 90-day loitering scenario accumulates **18.7 km** of position error (3σ), a 12.3× increase over the direct transit baseline. This error growth is driven primarily by maneuver execution errors and desaturation perturbations, which compound over multiple NRHO revolutions.

H.3 H.3 Conjunction Probability Analysis

The probability of conjunction P_c with the Gateway is calculated using a Gaussian error model:

$$P_c = 1 - \exp\left(-\frac{d^2}{2\sigma_{pos}^2}\right) \quad (6)$$

where d is the separation distance and σ_{pos} is the cumulative position error.

Phase Offset	Separation (km)	Position Error (km)	Conjunction Probability
60 min	400	18.7	0.11%
30 min	200	18.7	2.3%
10 min	50	18.7	68.4%

Table 34: Conjunction probability by phase offset

Analysis:

- **60-min offset (400 km):** $P_c = 0.11\%$ — **SAFE** but requires longer transfer times.
- **30-min offset (200 km):** $P_c = 2.3\%$ — **MARGINAL** (exceeds 1% threshold).
- **10-min offset (50 km):** $P_c = 68.4\%$ — **UNSAFE** (catastrophic risk).

H.4 H.4 Risk Mitigation: Direct Transit Mandate

To eliminate conjunction risk, the TUG-BIG architecture enforces a **Direct Transit Mandate**:

1. **Maximum Loitering:** Capped at **6.5 days** (1 NRHO revolution) for phasing only.
2. **Position Error Budget:** Limited to <5 km (3σ) to maintain $P_c < 0.5\%$ at 60-min offset.
3. **No Extended Loitering:** 90-day scenarios are **REJECTED** due to unacceptable conjunction probability.

H.5 H.5 Error Propagation CDR Verdict

Criterion	Assessment	Status
90-day position error	18.7 km (3σ) — exceeds safe margins	FAIL
Conjunction probability (60-min)	0.11% — acceptable but suboptimal	PASS
Conjunction probability (30-min)	2.3% — exceeds 1% threshold	FAIL
Direct transit error (1.57 days)	1.52 km (3σ) — negligible risk	PASS

Table 35: Error propagation verification

Conclusion: The orbital error propagation analysis confirms that 90-day NRHO loitering introduces **unacceptable conjunction risks** due to cumulative navigation and maneuver execution errors. The TUG-BIG architecture's direct transit paradigm (1.57 days) maintains position errors below 2 km, ensuring safe operations without requiring large phase offsets. This validates the architectural decision to reject extended loitering scenarios.